

Elimination Methods

Slide Courtesy: Ms. Raqeebir Rab

Naïve Gauss Elimination

Very close to Elimination of Unknowns

1. Forward Elimination
2. Backward Substitution

Naïve because we don't consider division by zero to be a possibility

Consider the following system of linear equations:

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3\end{aligned}$$

It can be written as: $[A][X] = [b]$; A is the coefficient Matrix, X is vector of unknown and b is the vector of constant.

Naïve Gauss Elimination

1. Forward Elimination of Unknowns

- Reduce the coefficient matrix $[A]$ to an upper triangular system
- Eliminate x_1 from the 2nd to n^{th} Equations.
- Eliminate x_2 from the 3rd to n^{th} Equations.
- Continue process until the n^{th} equation has only 1 Non-Zero coefficient

$$\begin{aligned} & \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \\ \rightarrow & \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & 0 & a''_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b'_2 \\ b''_3 \end{bmatrix} \end{aligned}$$

Suppose the first equation is your pivot row. To turn the coefficient matrix into an upper triangular form, all the other entries in the first column need to be zero. To do this, we calculate a multiplier by dividing the coefficient of the variable in the current row by the pivot coefficient. Then, we subtract the appropriate multiple of the pivot row from the elimination row.

Naïve Gauss Elimination

1. Forward Elimination of Unknowns

$$\begin{array}{lcl} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 & \dots\dots\dots(1) & \text{(Pivot row)} \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2 & \dots\dots\dots(2) & \text{(Elimination row)} \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3 & \dots\dots\dots(3) & \end{array} \quad \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

To eliminate x_1 from the second equation, we multiply the first equation by $-a_{21}/a_{11}$ and then add it to the second equation

$$\begin{array}{l} a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2 \\ a_{11}(-a_{21}/a_{11})x_1 + a_{12}(-a_{21}/a_{11})x_2 + a_{13}(-a_{21}/a_{11})x_3 = b_1(-a_{21}/a_{11}) \end{array}$$

$\downarrow$
Pivot

$$0 + (a_{22} - a_{12}(a_{21}/a_{11}))x_2 + (a_{23} - a_{13}(a_{21}/a_{11}))x_3 = b_2 - b_1(a_{21}/a_{11})$$

Let,

$$(a_{22} - a_{12}(a_{21}/a_{11})) = a'_{22} \qquad a'_{22}x_2 + a'_{23}x_3 = b'_2 \dots\dots\dots(2')$$

$$(a_{23} - a_{13}(a_{21}/a_{11})) = a'_{23}$$

$$b_2 - b_1(a_{21}/a_{11}) = b'_2 \qquad \text{Replace equation 2 by equation 2'}$$

As you can see, we use the pivot row to make changes to the elimination row, setting the coefficient of the variable in that row to zero

Naïve Gauss Elimination

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \dots\dots\dots(1) \text{ (Pivot row)}$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2 \dots\dots\dots(2)$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3 \dots\dots\dots(3) \text{ (Elimination row)}$$

Pivot Element



To eliminate x_1 from the second equation, we multiply the first equation by $-a_{31}/a_{11}$ and then add it to the third equation

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

$$a_{11}(-a_{31}/a_{11})x_1 + a_{12}(-a_{31}/a_{11})x_2 + a_{13}(-a_{31}/a_{11})x_3 = b_1(-a_{31}/a_{11})$$

$$0 + (a_{32} - a_{12}(a_{31}/a_{11}))x_2 + (a_{33} - a_{13}(a_{31}/a_{11}))x_3 = b_3 - b_1(a_{31}/a_{11})$$

Let,

$$(a_{32} - a_{12}(a_{31}/a_{11})) = a'_{22}$$

$$a'_{32}x_2 + a'_{33}x_3 = b'_3 \dots\dots\dots(3')$$

$$(a_{23} - a_{13}(a_{21}/a_{11})) = a'_{23}$$

$$b_2 - b_1(a_{21}/a_{11}) = b'_2$$

Replace equation 3 by equation 3'

Naïve Gauss Elimination

1. Forward Elimination of Unknowns

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \dots\dots\dots(1)$$

$$a'_{22}x_2 + a'_{23}x_3 = b'_2 \dots\dots\dots(2') \text{ (Pivot row)}$$

$$a'_{32}x_2 + a'_{33}x_3 = b'_3 \dots\dots\dots(3') \text{ (Elimination row)}$$

Now, to eliminate x_2 from the 3' equation, we multiply the 2' equation by $-a'_{32}/a'_{22}$ and then add it to the 3' equation.

$$a'_{32}x_2 + a'_{33}x_3 = b'_3$$

$$a'_{22}(-a'_{32}/a'_{22})x_2 + a'_{23}(-a'_{32}/a'_{22})x_3 = (-a'_{32}/a'_{22})b'_2$$

$$0 + (a'_{33} - a'_{23}(a'_{32}/a'_{22})) = b'_3 - (a'_{32}/a'_{22})b'_2$$

Let

$$a'_{33} - a'_{23}(a'_{32}/a'_{22}) = a''_{33}$$

$$a''_{33}x_3 = b''_3 \dots\dots\dots(3'')$$

$$b'_3 - (a'_{32}/a'_{22})b'_2 = b''_3$$

Naïve Gauss Elimination

1. Forward Elimination

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \dots\dots\dots(1)$$

$$a'_{22}x_2 + a'_{23}x_3 = b'_2 \dots\dots\dots(2')$$

$$a''_{33}x_3 = b''_3 \dots\dots\dots(3'')$$

Solve for x_3

Naïve Gauss Elimination

2. Backward Substitution

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \dots\dots\dots(1)$$

$$a'_{22}x_2 + a'_{23}x_3 = b'_2 \dots\dots\dots(2')$$

$$a''_{33}x_3 = b''_3 \dots\dots\dots(3'')$$

$$x_3 = \frac{b''_3}{a''_{33}}$$

From 2'

$$x_2 = \frac{b'_2 - a'_{23}x_3}{a'_{22}}$$

From 1

$$x_1 = \frac{b_1 - a_{13}x_3 - a_{12}x_2}{a_{11}}$$

Naïve Gauss Elimination

General Form

$$x_n = \frac{b_n^{n-1}}{a_{nn}^{n-1}}$$

$$x_i = \frac{b_i^{i-1} - \sum_{j=i+1}^n a_{ij}^{(i-1)} x_j}{a_{ii}^{i-1}}$$

Naïve Gauss Elimination

Example: Solve the following system using Naïve Gauss Elimination Method

$$x - 3y + z = 4 \dots\dots\dots(1)$$

$$2x - 8y + 8z = -2 \dots\dots\dots(2)$$

$$-6x + 3y - 15z = 9 \dots\dots\dots(3)$$

Naïve Gauss Elimination

Row 1: $x - 3y + z = 4$(1) (Pivot Row)

Row 2: $2x - 8y + 8z = -2$(2) (Elimination Row)

Row 3: $-6x + 3y - 15z = 9$(3)

Forward Elimination:

Step 1: Eliminate x from Row 2

Row 2 = Row 2 + $(-2)^*$ Row 1

$$2x - 8y + 8z = -2$$

$$-2x + 6y - 2z = -8$$

$$-2y + 6z = -10$$

$$\Rightarrow -y + 3z = -5 \text{(2')}$$

Naïve Gauss Elimination

Row 1: $x - 3y + z = 4$(1) (Pivot Row)

Row 2: $2x - 8y + 8z = -2$(2)

Row 3: $-6x + 3y - 15z = 9$(3) (Elimination Row)

Step 2: Eliminate x from Row 3

Row 3 = Row 3 + (6) * Row 1

$$-6x + 3y - 15z = 9$$

$$6x - 18y + 6z = 24$$

$$\begin{aligned} & -15y - 9z = 33 \\ \Rightarrow & -5y - 3z = 11 \text{(3')} \end{aligned}$$

Naïve Gauss Elimination

Row 1: $x - 3y + z = 4$(1)

Row 2': $-y + 3z = -5$(2') (Pivot Row)

Row 3': $-5y - 3z = 11$(3') (Elimination Row)

Step 3: Eliminate y from 3'

Row 3' = Row 3' + (-5) Row 2'

$$-5y - 3z = 11$$

$$5y - 15z = 25$$

$$-18z = 36$$
.....(3'')

Naïve Gauss Elimination

$$\text{Row 1: } x - 3y + z = 4 \dots\dots\dots (1)$$

$$\text{Row 2': } -y + 3z = -5 \dots\dots\dots (2')$$

$$\text{Row 3'': } -18z = 36 \dots\dots\dots (3'')$$

Back Substitution:

From (3'')

$$-18z = 36$$

$$z = -2$$

From (2')

$$-y + 3(-2) = -5$$

$$y = -1$$

From (1)

$$x - 3(-1) - 2 = 4$$

$$x = 3$$

$$\text{Solution: } x = 3 ; y = -1 ; z = -2$$

Gauss Elimination With Pivoting

Problem of Naïve Gauss Elimination:

During both the elimination and back substitution a division by zero may occur

Consider the following system:

$$\begin{aligned}2x_2 + 3x_3 &= 8 \\4x_1 + 6x_2 + 7x_3 &= -3 \\2x_1 - 3x_2 + 6x_3 &= 5\end{aligned}$$

The normalization of the first row would involve division by zero

If magnitude of pivot element is small compared to other elements, then round off errors may occur.

Solution:

The row with zero pivot element should be interchanged with the row having the largest (absolute) coefficient in that position

Gauss Elimination With Partial Pivoting

Steps of Gauss Elimination with Pivoting:

1. Considering the first row as the pivot row, search and locate the largest absolute value among the coefficients in the first column
2. Exchange the first row with the row containing that element.
3. Then eliminate the first variable in the other equations as explained earlier.
4. Continue this procedure till $(n-1)$ unknowns are eliminated.

This process is referred to as *partial pivoting*. There is an alternative scheme known as *complete pivoting* in which, at each stage, the largest element among the remaining coefficients is selected as the pivot.

Gauss Elimination With Partial Pivoting

Solve the following system using Gauss Elimination With Partial Pivoting.

$$2x_1 + x_2 + x_3 = 5$$

$$4x_1 - 6x_2 = -2$$

$$-2x_1 + 7x_2 + 2x_3 = 9$$

Gauss Elimination With Partial Pivoting

Step 1: Find the row with largest coefficient at 1st column

$$\text{Row 1: } 2x_1 + x_2 + x_3 = 5 \dots\dots\dots(1)$$

$$\text{Row 2: } 4x_1 - 6x_2 = -2 \dots\dots\dots(2)$$

$$\text{Row 3: } -2x_1 + 7x_2 + 2x_3 = 9 \dots\dots\dots(3)$$

Row 2 has largest coefficient so, interchange row 1 and row 2 and system become:

$$\text{Row 1: } 4x_1 - 6x_2 = -2 \dots\dots\dots(1)$$

$$\text{Row 2: } 2x_1 + x_2 + x_3 = 5 \dots\dots\dots(2)$$

$$\text{Row 3: } -2x_1 + 7x_2 + 2x_3 = 9 \dots\dots\dots(3)$$

Gauss Elimination With Partial Pivoting

$$\text{Row 1: } 4x_1 - 6x_2 = -2 \dots\dots\dots(1) \text{ (Pivot Row)}$$

$$\text{Row 2: } 2x_1 + x_2 + x_3 = 5 \dots\dots\dots(2) \text{ (Elimination row)}$$

$$\text{Row 3: } -2x_1 + 7x_2 + 2x_3 = 9 \dots\dots\dots(3)$$

Forward Elimination:

Step 2:

$$\text{Row 2} = \text{Row 2} + (-1/2) * \text{Row 1}$$

$$2x_1 + x_2 + x_3 = 5$$

$$-2x_1 + 3x_2 = 1$$

$$4x_2 + x_3 = 6 \dots\dots\dots(2')$$

Gauss Elimination With Partial Pivoting

$$\text{Row 1: } 4x_1 - 6x_2 = -2 \dots\dots\dots(1) \text{ (Pivot Row)}$$

$$\text{Row 2: } 2x_1 + x_2 + x_3 = 5 \dots\dots\dots(2)$$

$$\text{Row 3: } -2x_1 + 7x_2 + 2x_3 = 9 \dots\dots\dots(3) \text{ (Elimination row)}$$

Step 3:

$$\text{Row 3} = \text{Row 3} + (1/2) * \text{Row 1}$$

$$-2x_1 + 7x_2 + 2x_3 = 9$$

$$2x_1 - 3x_2 = -1$$

$$4x_2 + 2x_3 = 8 \dots\dots\dots(3')$$

Gauss Elimination With Partial Pivoting

$$\text{Row 1: } 4x_1 - 6x_2 = -2 \dots\dots\dots(1)$$

$$\text{Row 2': } 4x_2 + x_3 = 6 \dots\dots\dots(2')(\text{Pivot Row})$$

$$\text{Row 3': } 4x_2 + 2x_3 = 8 \dots\dots\dots(3') \text{ (Elimination row)}$$

Step 4:

$$\text{Row 3'} = \text{Row 3'} + (-1) * \text{Row 2'}$$

$$4x_2 + 2x_3 = 8$$

$$-4x_2 - x_3 = -6$$

$$x_3 = 2 \dots\dots\dots(3'')$$

Gauss Elimination With Partial Pivoting

Upper Triangular Matrix

$$\text{Row 1: } 4x_1 - 6x_2 = -2 \dots\dots\dots(1)$$

$$\text{Row 2': } 4x_2 + x_3 = 6 \dots\dots\dots(2') \Rightarrow$$

$$\text{Row 3'': } x_3 = 2 \dots\dots\dots(3'')$$

$$\begin{bmatrix} 4 & -6 & 0 & -2 \\ 0 & 4 & 1 & 6 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

Back Substitution:

From Row 3''

$$x_3 = 2$$

From Row 2'

$$4x_2 + 2 = 6$$

$$\Rightarrow x_2 = 1$$

From Row 1

$$4x_1 - 6 \cdot 1 = -2$$

$$\Rightarrow x_1 = 1$$

$$\text{Solution: } x_1 = 1; x_2 = 1; x_3 = -2$$

Complexity Analysis of Gauss Elimination Method

- Total Number of Flops (Floating point operations):
 - Total Number of multiplication/division $(n-1)(n+1)$
 - Total Number of addition/ subtraction $(n-1)(n)$
- Total Complexity:

$$\frac{2n^3}{2} + o(n^2) + n^2 + o(n)$$

Forward Elimination

Backward Substitution

$$\Rightarrow \frac{2n^3}{2} + o(n^2)$$

Gauss Jordan Method

- Another popular approach for solving system of linear equations.
- In Gauss-Jordan method, eliminates all the off-diagonal unknowns creating a Reduced Row Echelon Form (RREF) matrix. For a full rank square matrix, the RREF will be an identity matrix.
- All rows are normalized by dividing them by their pivot elements.
- Obtain the values of unknowns directly from the b vector, without employing back substitution

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$
$$\rightarrow \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a'_{22} & 0 \\ 0 & 0 & a''_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b'_2 \\ b''_3 \end{bmatrix}$$

Gauss Jordan Method

- Normalize the first equation by dividing it by its pivot element.
- Eliminate x_1 term from all the other equations.
- Now, normalize the second equation by dividing it by its pivot element.
- Eliminate x_2 from all the equations, above and below the normalized pivotal equation.
- Repeat this process until x_n is eliminated from all but the last equation.
- The resultant b vector is the solution vector.

Because all other elements in the pivot columns are 0, the solution can be read directly from the matrix, so no back-substitution is needed, unlike in Gauss Elimination.

Gauss Jordan Method

- Solve the following system of linear equations using Gauss-Jordan method

$$2x_1 + 4x_2 + 6x_3 = 18$$

$$4x_1 + 5x_2 + 6x_3 = 24$$

$$3x_1 + x_2 - 2x_3 = 4$$

You can show the operations step by step, like in Gauss Elimination, or present the calculations using a matrix. Either way, make sure to include all the detailed steps.

Gauss Jordan Method

Step	Equation Form	Augmented Matrix Form	Next Step
	$R1: 2x_1 + 4x_2 + 6x_3 = 18$ $R2: 4x_1 + 5x_2 + 6x_3 = 24$ $R3: 3x_1 + x_2 - 2x_3 = 4$	$\left[\begin{array}{ccc c} 2 & 4 & 6 & 18 \\ 4 & 5 & 6 & 24 \\ 3 & 1 & -2 & 4 \end{array} \right]$	Normalize R1 $R1 \rightarrow 1/2 R1$
1	$R1: x_1 + 2x_2 + 3x_3 = 9$ $R2: 4x_1 + 5x_2 + 6x_3 = 24$ $R3: 3x_1 + x_2 - 2x_3 = 4$	$\left[\begin{array}{ccc c} 1 & 2 & 3 & 9 \\ 4 & 5 & 6 & 24 \\ 3 & 1 & -2 & 4 \end{array} \right]$	Eliminate x_1 from R1 and R2 $R2 \rightarrow R2 - 4R1$ $R3 \rightarrow R3 - 3R1$
2	$R1: x_1 + 2x_2 + 3x_3 = 9$ $R2: -3x_2 - 6x_3 = -12$ $R3: -5x_2 - 11x_3 = -23$	$\left[\begin{array}{ccc c} 1 & 2 & 3 & 9 \\ 0 & -3 & -6 & -12 \\ 0 & -5 & -11 & -23 \end{array} \right]$	Normalize R2 $R2 \rightarrow -1/3 R2$

Gauss Jordan Method

Step	Equation Form	Augmented Matrix Form	Next Step
3	R1: $x_1 + 2x_2 + 3x_3 = 9$ R2: $x_2 + 2x_3 = 4$ R3: $-5x_2 - 11x_3 = -23$	$\begin{bmatrix} 1 & 2 & 3 & 9 \\ 0 & 1 & 2 & 4 \\ 0 & -5 & -11 & -23 \end{bmatrix}$	Eliminate x_2 from R1 and R2 $R1 \rightarrow R1 - (2) R2$ $R3 \rightarrow R3 - (-5) R2$
4	R1: $x_1 - x_3 = 1$ R2: $x_2 + 2x_3 = 4$ R3: $-x_3 = -3$	$\begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & -1 & -3 \end{bmatrix}$	Eliminate x_3 from R1 and R2 $R2 \rightarrow R2 - 2 R3$ $R1 \rightarrow R1 - (-1)R3$
6	R1: $x_1 = 4$ R2: $x_2 = -2$ R3: $-x_3 = -3$	$\begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & -1 & -3 \end{bmatrix}$	

Gauss Jordan Method

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \\ -3 \end{bmatrix}$$

Solution for the system:

$$x_1 = 4$$

$$x_2 = -2$$

$$x_3 = 3$$

Gauss Jordan Method

- The Gauss-Jordan method requires approximately 50 percent more arithmetic operations compared to Gauss method. Therefore, this method is rarely used.
- See following table: for the comparison of computational effort.

	Gauss Method	Gauss-Jordan Method
Multiplication	$\frac{1}{3} n^3$	$\frac{1}{2} n^3$
Subtraction	$\frac{1}{3} n^3$	$\frac{1}{2} n^3$
Divisions	$\frac{1}{2} n^2$	$\frac{1}{2} n^2$